

LAB # 5

DECISION MAKING IN PROGRAMMING USING SWITCH STATEMENT

Lab Tasks

1. What would be the output of following program?

```
a) main( )
{
    int i = 2 ;
    switch ( i )
    {
        case 1 :
            cout<<"I am in case 1 \n"<<endl;

        case 2 :
            cout<<"I am in case 2 \n"<<endl;

        case 3 :
            cout<<"I am in case 3 \n"<<endl;

        default :
            cout<<"I am in default \n"<<endl;
    }
}
```

2. Write a program to create Simple Calculator using switch case.
3. Write a program print total number of days in a month using switch case.
4. Develop a four basic mathematical functions calculator that takes input from user and performs any one of the selected operations and displays the result. Use switch case approach to develop this program. Also make use of “goto” statement to continuously take input of the operator, if user enters some unknown operator (other than, +, -, *, /)
5. Write a Program to take the value from the user as input all sides of a triangle and check whether the triangle is valid or not. Using switch statement

6. Using switch statement Write a program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer. Calculate percentage and grade according to following:

Percentage $\geq 90\%$: Grade A

Percentage $\geq 80\%$: Grade B

Percentage $\geq 70\%$: Grade C

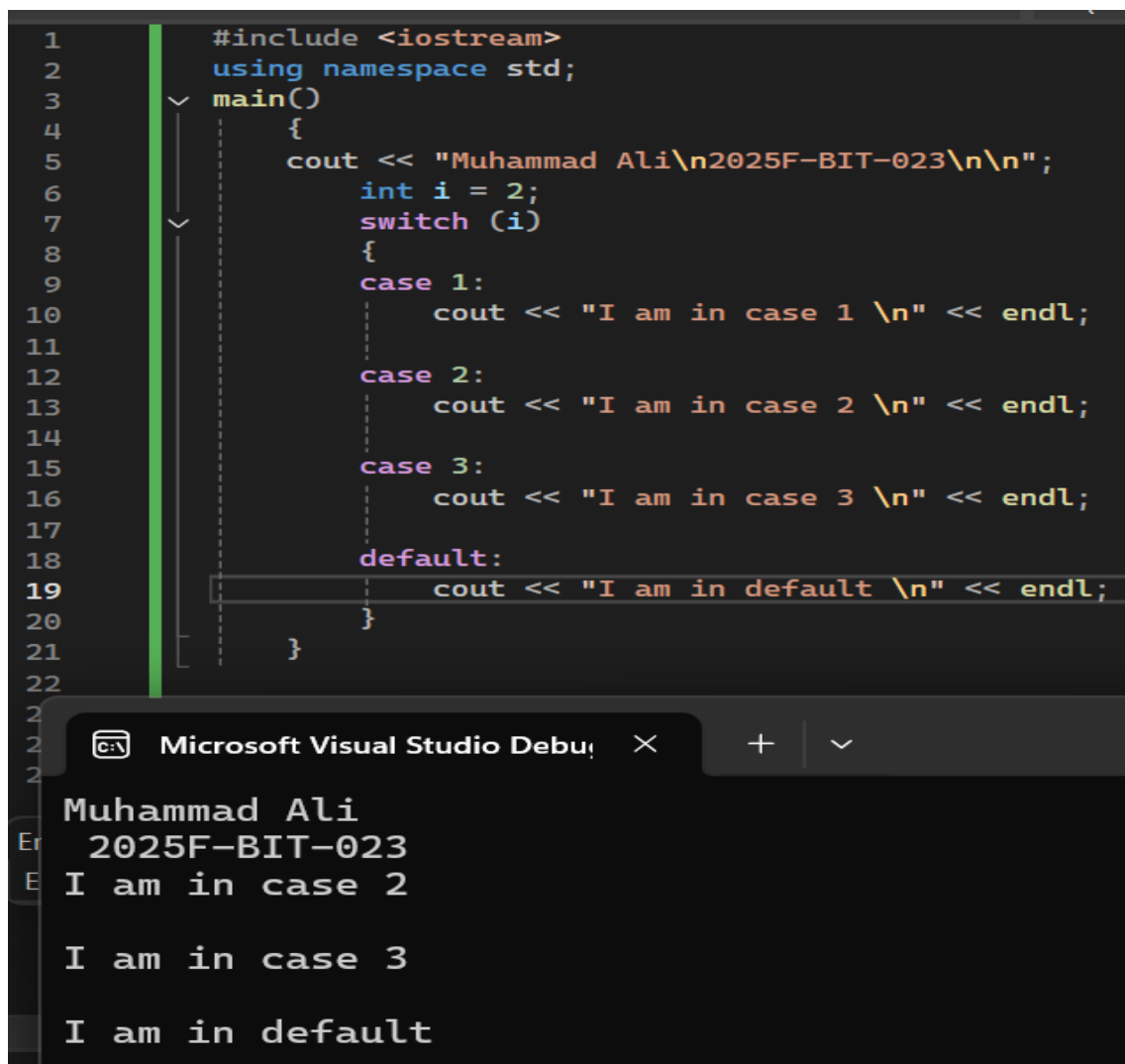
Percentage $\geq 60\%$: Grade D

Percentage $\geq 40\%$: Grade E

Percentage $< 40\%$: Grade F

1. What would be the output of following program?

```
a) main( )
{
    int i = 2;
    switch ( i )
    {
        case 1 :
            cout<<"I am in case 1 \n"<<endl;
        case 2 :
            cout<<"I am in case 2 \n"<<endl;
        case 3 :
            cout<<"I am in case 3 \n"<<endl;
        default :
            cout<<"I am in default \n"<<endl;
    }
}
```



The screenshot shows a C++ program in Visual Studio. The code defines a `main()` function that prints "Muhammad Ali\n2025F-BIT-023\n\n" and then enters a `switch` statement with `i = 2`. The `switch` has cases for 1, 2, 3, and a default. Since `i` is 2, the program falls through to the default case. The output window shows the program's execution results.

```
1  #include <iostream>
2  using namespace std;
3  main()
4  {
5      cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6      int i = 2;
7      switch (i)
8      {
9          case 1:
10             cout << "I am in case 1 \n" << endl;
11          case 2:
12             cout << "I am in case 2 \n" << endl;
13          case 3:
14             cout << "I am in case 3 \n" << endl;
15          default:
16             cout << "I am in default \n" << endl;
17      }
18  }
```

Microsoft Visual Studio Debug Console Output:

```
Muhammad Ali
2025F-BIT-023
I am in case 2
I am in case 3
I am in default
```

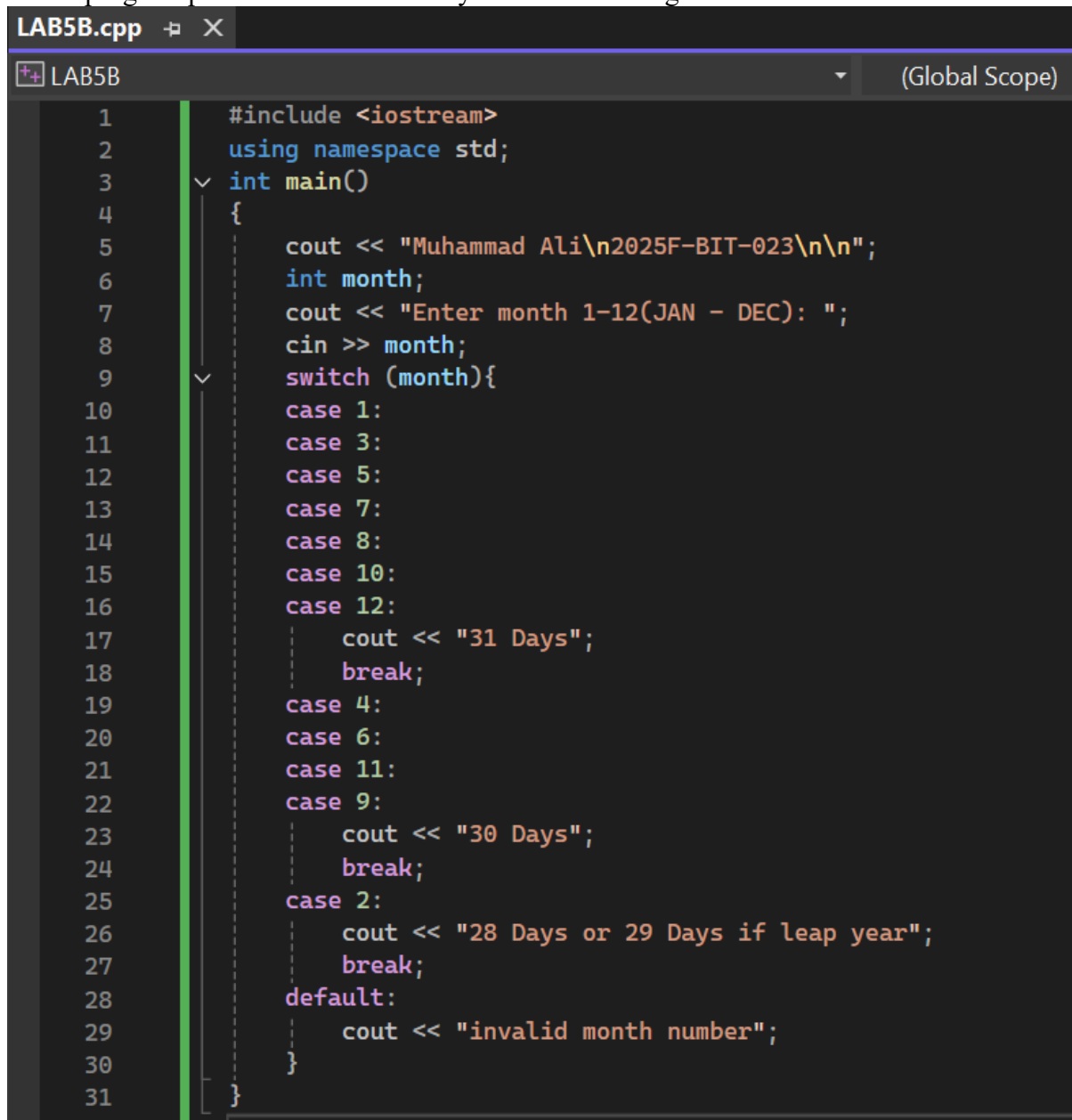
2. Write a program to create Simple Calculator using switch case.

```
lab5 (Global Scope)
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6      char opr;
7      int num1,num2;
8      cout << "Enter first number :";
9      cin >> num1;
10     cout << "Enter second number :";
11     cin >> num2;
12     cout << "Enter operation + - * / : ";
13     cin >> opr;
14     switch (opr)
15     {
16     case '+':
17         cout << num1 << "+" << num2 << "=" << num1 + num2 << endl;
18         break;
19     case '-':
20         cout << num1 << "-" << num2 << "=" << num1 - num2 << endl;
21         break;
22     case '*':
23         cout << num1 << "*" << num2 << "=" << num1 * num2 << endl;
24         break;
25     case '/':
26         cout << num1 << "/" << num2 << "=" << num1 / num2 << endl;
27         break;
28     default:
29         cout << "invalid operator\n" << endl;
30     }
31 }
```

```
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023

Enter first number :3
Enter second number :8
Enter operation + - * / : *
3*8=24
```

3. Write a program print total number of days in a month using switch case.



```
LAB5B.cpp  X
LAB5B (Global Scope)
1      #include <iostream>
2      using namespace std;
3      int main()
4      {
5          cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6          int month;
7          cout << "Enter month 1-12(JAN - DEC): ";
8          cin >> month;
9          switch (month){
10         case 1:
11         case 3:
12         case 5:
13         case 7:
14         case 8:
15         case 10:
16         case 12:
17             cout << "31 Days";
18             break;
19         case 4:
20         case 6:
21         case 11:
22         case 9:
23             cout << "30 Days";
24             break;
25         case 2:
26             cout << "28 Days or 29 Days if leap year";
27             break;
28         default:
29             cout << "invalid month number";
30         }
31     }
```



```
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023

Enter month 1-12(JAN - DEC): 4
30 Days

C:\Users\DELL\source\repos\LAB5F\vc14\Debug\LAB5F.exe (process 20000) exited with code 0
```

4. Develop a four basic mathematical functions calculator that takes input from user and performs any one of the selected operations and displays the result. Use switch case approach to develop this program. Also make use of “goto” statement to continuously take input of the operator, if user enters some unknown operator (other than, +, -, *, /)

```

#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    start:
    int n1, n2;
    char optr;
    cout << "Enter your operation: \n1)+ for addition\n2)- for subtraction\n3)* for multiplication\n4)/ for division \n";
    cin >> optr;
    cout << "Enter first number : ";
    cin >> n1;
    cout << "Enter second number : ";
    cin >> n2;
    switch (optr) {
        case '+':
            cout << "The addition of given numbers is : " << n1 + n2;
            break;
        case '-':
            cout << "The subtraction of given numbers is : " << n1 - n2;
            break;
        case '*':
            cout << "The multiplication of given numbers is : " << n1 * n2;
            break;
        case '/':
            cout << "The division of given numbers is : " << n1 / n2;
            break;
        default:
            cout << "invalid operator\n\n";
            goto start;
    }
}

```

Microsoft Visual Studio Debug Console Output:

```

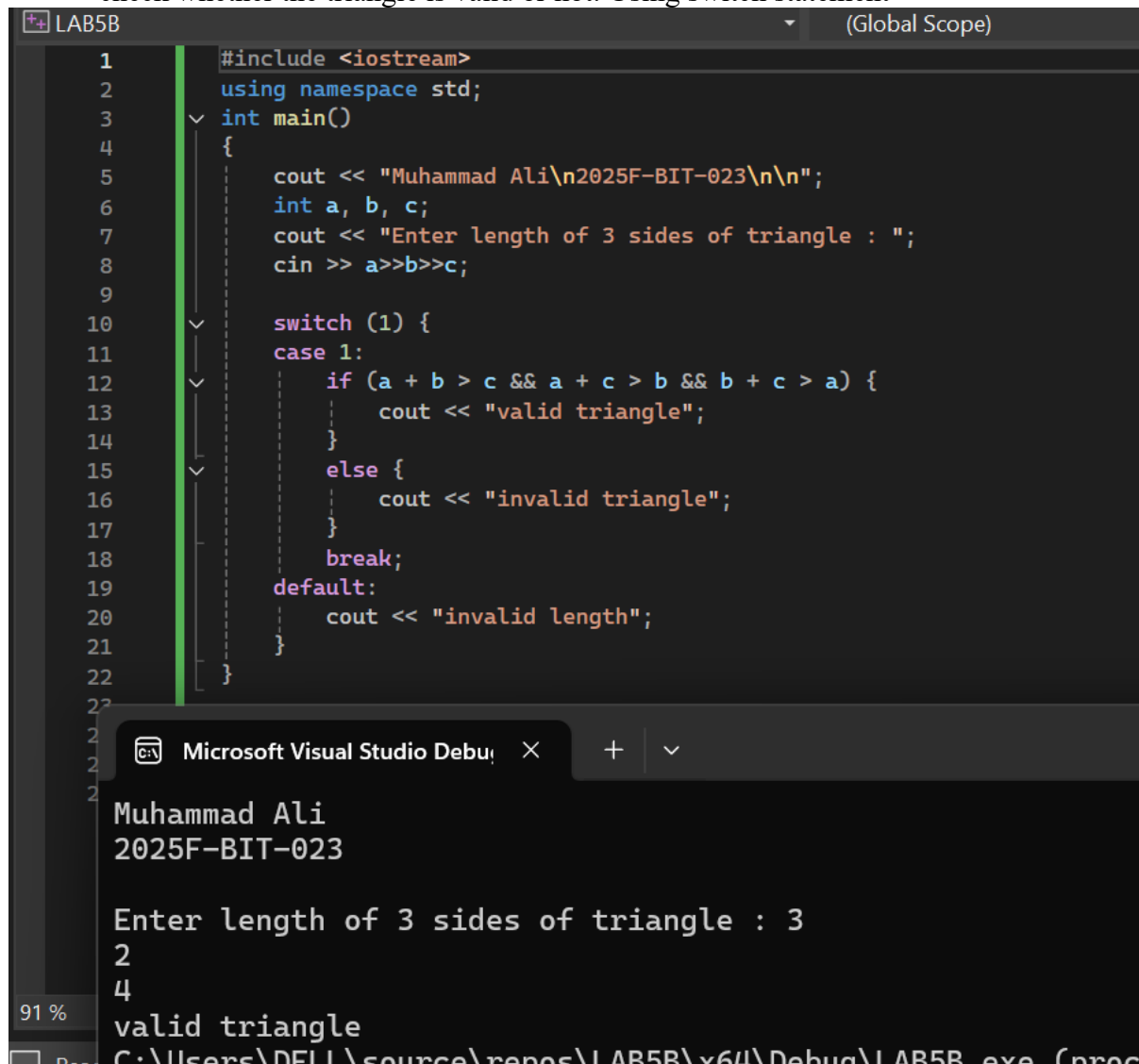
Muhammad Ali
2025F-BIT-023

Enter your operation:
1)+ for addition
2)- for subtraction
3)* for multiplication
4)/ for division
2
Enter first number : 2
Enter second number : 2
invalid operator

Enter your operation:
1)+ for addition
2)- for subtraction
3)* for multiplication
4)/ for division
*
Enter first number : 3
Enter second number : 2
The multiplication of given numbers is : 6

```

5. Write a Program to take the value from the user as input all sides of a triangle and check whether the triangle is valid or not. Using switch statement



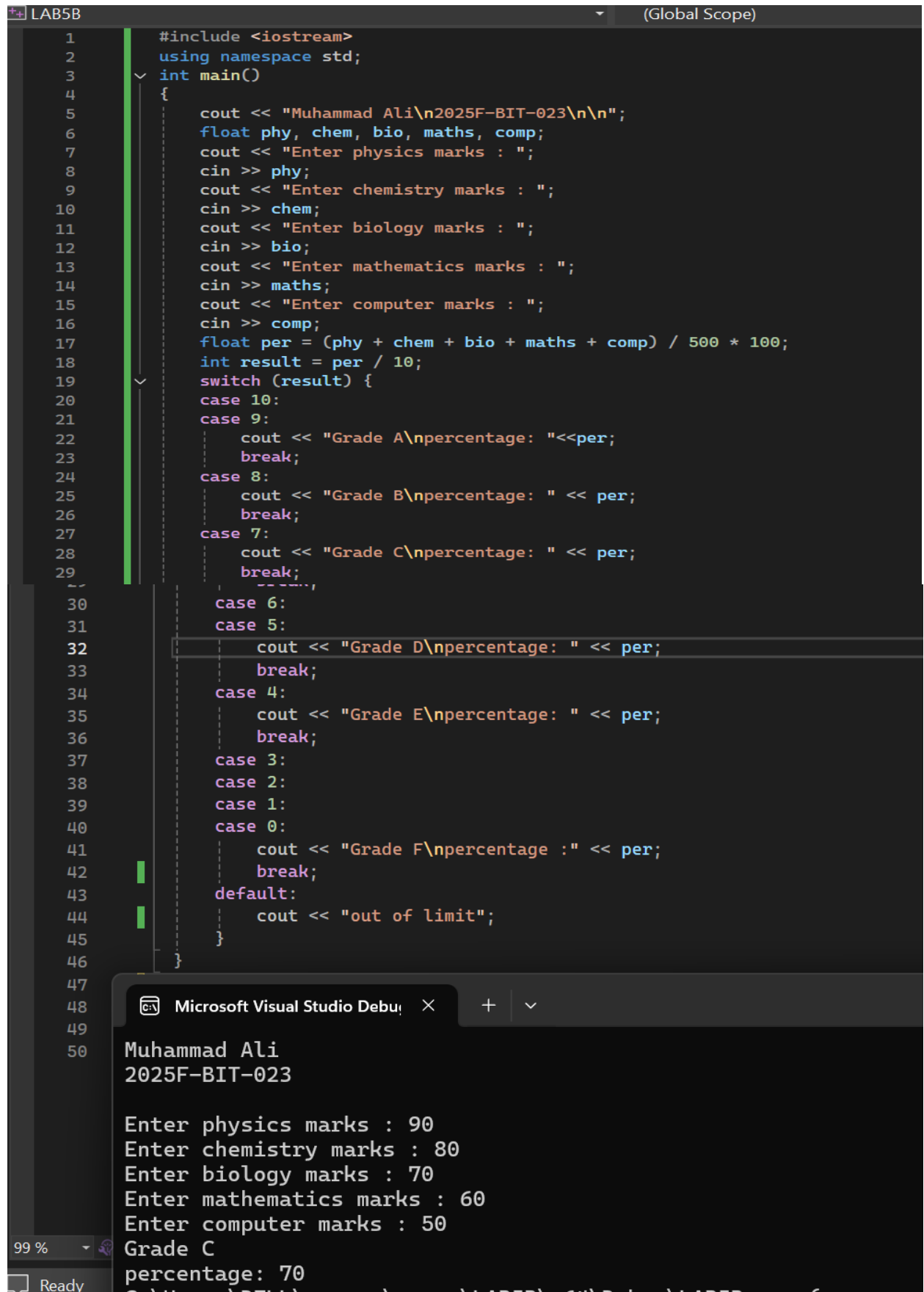
```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6      int a, b, c;
7      cout << "Enter length of 3 sides of triangle : ";
8      cin >> a>>b>>c;
9
10     switch (1) {
11     case 1:
12         if (a + b > c && a + c > b && b + c > a) {
13             cout << "valid triangle";
14         }
15         else {
16             cout << "invalid triangle";
17         }
18         break;
19     default:
20         cout << "invalid length";
21     }
22 }
```

Microsoft Visual Studio Debug Console Output:

```
Muhammad Ali
2025F-BIT-023

Enter length of 3 sides of triangle : 3
2
4
valid triangle
```

6. Using switch statement Write a program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer. Calculate percentage and grade according to following:
- Percentage $\geq 90\%$: Grade A
 - Percentage $\geq 80\%$: Grade B
 - Percentage $\geq 70\%$: Grade C
 - Percentage $\geq 60\%$: Grade D
 - Percentage $\geq 40\%$: Grade E
 - Percentage $< 40\%$: Grade F



```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6     float phy, chem, bio, maths, comp;
7     cout << "Enter physics marks : ";
8     cin >> phy;
9     cout << "Enter chemistry marks : ";
10    cin >> chem;
11    cout << "Enter biology marks : ";
12    cin >> bio;
13    cout << "Enter mathematics marks : ";
14    cin >> maths;
15    cout << "Enter computer marks : ";
16    cin >> comp;
17    float per = (phy + chem + bio + maths + comp) / 500 * 100;
18    int result = per / 10;
19    switch (result) {
20    case 10:
21    case 9:
22        cout << "Grade A\npercentage: " << per;
23        break;
24    case 8:
25        cout << "Grade B\npercentage: " << per;
26        break;
27    case 7:
28        cout << "Grade C\npercentage: " << per;
29        break;
30    case 6:
31    case 5:
32        cout << "Grade D\npercentage: " << per;
33        break;
34    case 4:
35        cout << "Grade E\npercentage: " << per;
36        break;
37    case 3:
38    case 2:
39    case 1:
40    case 0:
41        cout << "Grade F\npercentage: " << per;
42        break;
43    default:
44        cout << "out of limit";
45    }
46 }
```

Microsoft Visual Studio Debug Console Output:

```
Muhammad Ali
2025F-BIT-023

Enter physics marks : 90
Enter chemistry marks : 80
Enter biology marks : 70
Enter mathematics marks : 60
Enter computer marks : 50
Grade C
percentage: 70
```